

B Homework: QED (due Thu 12th Mar 5pm)

B.1 Feynman diagrams

(a) Using only the QED vertex (4.4), draw the lowest order (in α_{QED}) Feynman diagrams for:

- (i) $e^-e^+ \rightarrow e^-e^+$,
- (ii) $e^-\mu^+ \rightarrow e^-\mu^+$,
- (iii) $\gamma\gamma \rightarrow \gamma\gamma$,
- (iv) $\gamma p \rightarrow e^-e^+p$ treating the proton p as a point particle

State how many powers of α_{EM} are in the cross-section for each process.

(b) Why can $e^-e^+ \rightarrow \mu^-\mu^+$ not proceed via a t -channel diagram?

(c) Given the range R of a massive force carrier is $R = 1/m$, restore $\hbar c$ to numerically estimate R for a pion ($m_\pi \approx 140$ MeV) and W boson ($m_W \approx 80$ GeV).

B.2 Gauge theory

Suppose we rescaled the phase in equation (4.46) to a general charge $Q\beta(x)$ such that $\psi(x) \rightarrow \psi'(x) = \psi(x)e^{iQ\beta(x)}$. Inserting this into the Dirac equation with an appropriately adjusted covariant derivative and following similar steps to arrive at (4.50), show what the transformation $A^\mu \rightarrow A'^\mu$ must be to restore invariance of the Dirac equation.

B.3 Feynman rules to cross-section

(a) Using the Feynman rules in section 4.4 (or textbook), write down the amplitude of the four Feynman diagrams sketched in figure 25: (i) $e^-e^- \rightarrow e^-e^-$, (ii) $e^-e^+ \rightarrow \gamma\gamma$, (iii) $e^-\gamma \rightarrow e^-\gamma$, (iv) $\gamma\gamma \rightarrow e^-e^+$. Draw the diagrams and label the parts that correspond to mathematical expressions for the spinors u, v , polarisation vectors ϵ_μ , and propagators.

(b) Considering ultra-relativistic $e^-e^+ \rightarrow \mu^-\mu^+$ scattering for the non-zero helicity combinations (6.7), sketch the arrows denoting the momentum $\mathbf{p}$ and helicity $\mathbf{s}$ vectors for the ingoing electron and positron, and outgoing muon and anti-muon.

(c) For each of the four possible helicity combinations, sketch graphs of $\mathcal{A}$ vs. $\cos\theta$ based on the amplitude dependence $\propto (1 \pm \cos\theta)^2$ from eq. (6.20).

(d) Starting from the differential cross-section $d\sigma = \frac{1}{\Phi} |\mathcal{M}|^2 dn_{2\text{-body}}$ with the incident flux $\Phi = 2s$, the averaged sum over the possible spin states $|\mathcal{M}|^2 = \frac{1}{4} \sum_{\text{spins}} |\mathcal{A}_{\text{spins}}|^2$ from the lecture notes eq. (6.21), and the 2-body Lorentz invariant phase space $dn_{2\text{-body}} = \frac{p_f}{E_{\text{CM}}} \frac{d\Omega}{(4\pi)^2}$ from eq. (5.38), show the steps to obtain $\sigma = \frac{4}{3} \frac{\pi \alpha_{\text{EM}}^2}{s}$.